Node Strings: Protein Folds Read Back to Operands on the Native Binary Rings

Water, the bond, the node string, the pair word and the open end, run on 30 training structures and 155 E. coli chains

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Abstract

The NEXUS fold compiler writes a fold forward from operands on the native binary rings. This paper reads finished folds back. Each noun in a solved structure is taken back to the transformation that made it, and that result back again, until only operators and their operands remain in a string. Water's open slots, in every order, are words whose partner is their complement, and promotion carries that facing to every lane above. On the two-bit ring the two partner pairs are the two bonds: opposite spins under reversal, a full orbital facing an empty one under complement. In 155 *E. coli* chains the backbone closes 62 percent of its own N–H slots, the side chains 15 percent and ordered water 10 percent, and each chain edge a water carries lowers its B-factor one step, from 0.99 to 0.74 of the file median. The backbone closures reduce to node strings, three ring operators placed on the chain (successor, facing and shift), about nine per hundred residues with five closures each and 78 percent of all closures inside a node. Each closure is one word of a pair ring, W(i, j) = b(i) ‖ J(b(j)): the facing law swaps its operands, the ring's halves carry its direction, and its two centre bits hold the parity sector that every register slip flips, on 239 of 239 odd steps and on none of 5,230 even ones. Inside a node the count is exact. The end is not in the node: a binary carry does not mark it, the field's criteria gather at 94 percent of ends and at 61 percent of the steps where a node went on, and facing stretches survive each pleat step at a steady rate with no wall. On the genetic code the same ring words sit at the same location: the R4 boxes that every operator pairs alike are glycine and proline, the compiler's lookahead pair, and the boxes of the start codon and two of the stops. The end is the open slot the emitter must be given.

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Every number here is printed by the reproduction notebook from the shipped data, the new results by its section 14.

1. Reading back to operands

The NEXUS fold compiler holds that transformations constitute what physically exists and that verbs precede nouns. A protein is not an object that later gets folded; it is a deterministic execution route through a pre-existing relational memory map, and the fold is the readout that route leaves behind (Kulik 2026a). The compiler runs the route forward. The sequence is the code that runs, the partial instruction 2 + = that fixes the operator and the first operand, and the field (water, pH, the ribosome's stream, the chaperones) is the code that runs the code, supplying the second operand at each step.

A compiled program can also be read back. A decompiler starts from the finished binary and follows each instruction to the operation that wrote it. This paper runs the compiler in that direction on solved folds. Every noun in a fold is the result of a transformation: a helix, a sheet, a bound water, a hydrogen bond. Each is taken back to the operation that made it, and the result of that operation back again, recursively, until the edge is all verbs and only operands in a string remain. The read-back is node-based, not linear. It does not walk the chain residue by residue comparing candidates; it places operators on the chain and lets each operator compute what it covers. What no operator covers is the part the read-back cannot own, and that part is the specification for the emitter that will write folds forward.

The construction is dependency injection. The amino acids are the constructor and the field is injected, and once injected the field is everywhere the chain is. So the read-back does not stop at the chain. Water is decompiled with it, together with what holds a water in place until an edge stops it from rolling out.

The path of the paper is the path the compiler set. A fold is first a list of closures. Water and the bond come first, because the field is injected before the chain closes on anything (section 2), and the census of what closes each backbone N–H shows where the field takes over from the chain (section 3). Taking the closures back to the operators that compute them gives the node string (section 4). Taking each closure back to its two operands gives the pair word, whose algebra carries the closure's facing, direction, count and parity sector (section 5). One piece does not reduce to the node's own operands: its end (section 6). The genetic code holds the same ring words at the same location (section 7), and the ledger and the emitter's specification close the paper (sections 8 and 9).

The structures are ones the program has already read: the 30 training structures of the compiler paper (2,656 residues) and the 155 *E. coli* chains built from exact genes (16,877 residues), with the crystal waters of their entries in the Protein Data Bank (Berman et al. 2000). A route back on data already seen names owners; it is not a landing. Statuses follow the compiler's ledger. RUNS marks what the ring algebra computes on every word and what the decompiler computes from the shipped data; STANDS marks results that established theory supplies; OPEN marks a missing link that is named but not yet written; DEAD END marks code that ran and landed where the landing belongs to another owner. The landing proper, a sealed prediction on chains the program has not read, belongs to the emitter. Every number below is printed by the reproduction notebook beside the value printed here: the new results by section 14, and the values carried from the compiler paper and the reading-direction note by the sections that reproduce those documents.

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2. Water and the bond as operands

2.1 The words that face their complement

The native partner P is reversal J on every word that reversal moves and complement C on the palindromes that reversal cannot move (Kulik 2026a). P and C agree on two classes of words: the palindromes, where P is C by construction, and the anti-palindromes, whose reversal is their complement. On these **P = C words** a word's partner is its complement.

The compiler paper writes water in the slot alphabet as 11110000: two polar hydrogens, 11 and 11, and two lone pairs, 00 and 00, every slot open. All six orderings of the four slots are P = C words. 00001111, 00110011, 11001100 and 11110000 are anti-palindromes; 00111100 and 11000011 are palindromes. None of the eight atom bytes in the compiler paper's table is a P = C word: amide N, α-carbon, carbonyl C, carbonyl O, hydroxyl O, carboxylate O^-, NH_3^+ and CH_2 all have partners that are not their complements. In whatever order a reader takes water's slots, water faces its complement, and no atom of the chain does.

Promotion carries water up the lanes. Promotion E(x) = x ‖ P(x) packs a word together with its partner and hands the pair up one lane as one word, and $P_{2n}(E(x)) = E(C_n(x))$ for every word (Kulik 2026b). On a P = C word P(x) = C(x), and P undoes itself, so P(C(x)) = x and

$$P(E(x)) = E(C(x)) = C(x) \,\|\, P(C(x)) = C(x) \,\|\, x = C(E(x)),$$

and the promoted word is again a P = C word. Water's 11110000 promotes to the palindrome 1111000000001111, which promotes to an anti-palindrome, and on up, alternating lane after lane. This is the injected field in ring terms: once water is in, its facing is inherited by every lane above it. **RUNS** on every P = C word of R1 through R10 and on four lanes of water.

2.2 The bond on the two-bit ring

Read each R2 word as the occupancy of one orbital: 00 empty, 10 one electron with spin up, 01 one electron with spin down, 11 full. P_2 pairs spin up with spin down, by reversal, and full with empty, by complement. The reversal pair is the covalent bond, two electrons of opposite spin in one shared orbital (Lewis 1916; Heitler & London 1927). The complement pair is the hydrogen bond read as donation: a full lone pair facing the empty antibonding orbital behind a polar hydrogen, n → σ* (Reed, Curtiss & Weinhold 1988). Two full orbitals, two empty ones, or two electrons of the same spin are not partners. Chemistry's two bonds are the smallest ring's two partner pairs at a different location, and P_2 is the bond rule. **STANDS**: established theory supplies both pairs.

The compiler paper's slot alphabet is this alphabet read through complement on the open slots. There a lone pair is 00 and a polar hydrogen 11; here the lone pair is the full orbital 11, and the polar hydrogen's open slot is the empty σ* orbital 00. Complement commutes with P_2 on all four words, so the two alphabets face the same way. In either alphabet water carries two slots of each kind: two donor ends and two acceptor ends of hydrogen bonds. The P = C words are the words that face by complement, and on R2 complement is the hydrogen bond: water is the operand that faces by hydrogen bond on every lane.

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3. What holds in the water

Water is read back by what closes on it. Each backbone N–H of the *E. coli* chains is read for its partner in the compiler's order: the chain's own backbone (a backbone O within 3.5 Å, more than two residues away, the compiler paper's closure rule), else a side-chain oxygen of the chain within 3.5 Å, else an ordered crystal water within 3.5 Å, else nothing in the model. Prolines, which carry no N–H, and each chain's first residue are skipped. Of the 155 chains, 154 carry crystal waters. Table 1 and Figure 2a give the census by the state each residue carries in the working set.

Table 1. What closes each backbone N–H in the 154 *E. coli* chains with crystal waters. Shares of the slots in each class.

N–H class	N–H slots	Chain backbone	Side chain	Ordered water	No partner in the model
All	15,880	0.623	0.147	0.101	0.129
Helix	5,940	0.771	0.121	0.056	0.051
Strand	5,683	0.547	0.143	0.133	0.177
Coil	4,257	0.519	0.188	0.122	0.171
First four N–H of a helix	1,828	0.583	0.215	0.103	0.098
β seam, register kept	3,210	0.775	0.069	0.073	0.083
β seam, slip residue	160	0.719	0.119	0.062	0.100

The chain closes 0.77 of its own N–H slots, 0.62 on its backbone and 0.15 on its side chains, and ordered water closes 0.10. The remaining 0.13 have no partner in the model. They are unlinked, not open: a crystal model carries only the water it orders, so these slots are not counted as open slots of the fold.

The states close differently. A helix is self-contained: its backbone closes 0.77 of its N–H, its side chains 0.12 and water 0.06, and 0.05 are unlinked. A strand hands more to the field: its backbone closes 0.55, water 0.13, and 0.18 are unlinked, the outward N–H of edge strands. Coil leans on side chains, at 0.19.

The first four N–H of a helix have no carbonyl four residues back inside the helix. Their backbone share falls to 0.58, and side chains take the slot at 0.21, nearly twice the helix rate, with water at 0.10. This is the N-cap, where serine, threonine, aspartate and asparagine close the start of a helix (Richardson & Richardson 1988; Aurora & Rose 1998), read as an address: the successor operator does not reach its own first four slots, and the side chains do.

A register slip in a β seam is not a hole. At the 160 N–H of slip residues, where the crease sum moves by one, the chain closes 0.84 (0.72 on the backbone, 0.12 on side chains), the same share as in the kept register (0.78 and 0.07). The side chains take up what the backbone gives up at the slip, 0.12 against 0.07, and water does not flood in, 0.06 against 0.07. The slip re-registers the seam (Kulik 2026b); it does not open it.

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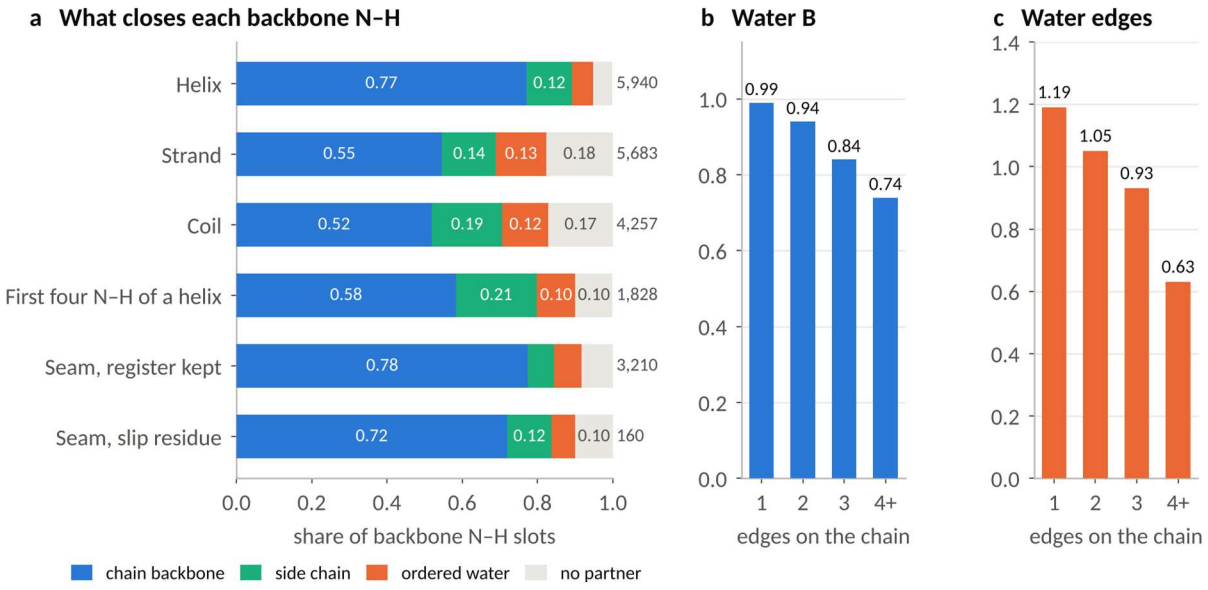


Figure 2. What holds in the water. (a) What closes each backbone N–H of the 154 *E. coli* chains with crystal waters, by state: the chain's backbone, a side-chain oxygen of the chain, an ordered water, or no partner in the model; slot counts at right. (b) Median B-factor of the waters that touch the chain, over each file's median water B, by the number of chain edges. (c) Mean edges from the same waters to other waters.

From the water's side, 11,572 waters touch a chain, with a chain nitrogen or oxygen within 3.5 Å: 6,805 by one edge, 3,130 by two, 1,134 by three and 503 by four or more. Each chain edge holds the water harder (Figure 2b, c). The median B-factor, over the file's median water B, falls from 0.99 at one edge to 0.94, 0.84 and 0.74, and the water's edges to other waters fall from 1.19 to 1.05, 0.93 and 0.63. A water with one chain edge moves like the median water: it is still rolling. Each further edge stops it by one more step, and a water with four chain edges has traded its water edges for chain edges and moves a quarter less than the median. It still moves. The hold is a gradient, one step per edge, not a switch; every closure the chain makes on a water takes one step off its motion. Crystallography sees the same gradient as the ordered first hydration layer (Nakasako 2004).

Of the waters that touch a chain, 4,015 bridge two of its residues. 2,121 join residues within five of each other, the reach of a successor node, and 1,894 join residues farther apart along the chain, the reach of a facing or shift node. Water writes closures of both kinds the chain writes. **RUNS** for the census and the hold.

4. Folds as node strings

With water and the bond read back, the backbone's own closures remain. A closure is the N–H of residue i closing on the O of residue j: the nearest backbone O within 3.5 Å, more than two residues away. Read as a list, a fold is nouns, a helix here and a sheet there. Read as operators, the list collapses, because most closures are computed from their neighbours by one of three ring operators placed on the chain.

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Tk, successor. The N–H of i closes on the O of i − k, with k = 3 or 4, over a run of donors with gaps of at most two. At k = 4 this is the α-helix (Pauling, Corey & Branson 1951); at k = 3, the 3_{10} helix and the turn. The partner is the successor count run back by k.

PS, facing. Residue i pairs with j = S − i across a crease, over a run of pairs that step one to three residues inward on both strands: the antiparallel ladder (Pauling & Corey 1951). The partner is the reflection about the crease at S/2. The ring's facing law reflects a word about the ring's crease; the ladder reflects a residue about the hairpin's. It is the same path at a different location.

Dd, shift. Residue i pairs with j = i + d, over a run that steps one to three residues forward on both strands: the parallel ladder.

A closure that no node covers is a **pointer**: its partner is written by address, not computed. A facing node can re-phase, moving its crease sum to S ± 1 inside the node, and keeps its pairs. Inside a node the partner is a route: the offset is known, and each partner is reached in one step from the last. A pointer is a closure whose offset no node supplies.

> **Reading a node string**
>
> The letter is the operator and the number its parameter: k for a successor node, the crease sum S for a facing node, the shift d for a shift node. The bracket gives the span of the first operand along the chain, residues counted from 0 along the model chain. A facing node that re-phases carries ~±1 or ~±2, the phases it reaches. Example: P16[0-4] pairs residue 0 with 16, 2 with 14 and 4 with 12.

The 30 training structures decompile to 243 nodes: 169 successor, 57 facing and 17 shift. They hold 1,195 of the 1,553 closures (0.77), 4.9 closures per node, at 9.1 nodes per 100 residues; the other 358 closures are pointers. The 155 *E. coli* chains decompile to 1,548 nodes: 925 successor, 392 facing and 231 shift. They hold 7,792 of the 9,983 closures (0.78), 5.0 per node, at 9.2 nodes per 100 residues, with 2,191 pointers. The two sets read back at the same rates. A fold is about nine operators per hundred residues. **RUNS.**

Ubiquitin's 44 closures read back to eight nodes:

```
P16[0-4] D62[1-5] D63[3-5] P74[18-23]~+1 T4[25-34] T3[39-40] P110[39-43] T3[57-59]
```

The β1–β2 hairpin is P16. The parallel pairing of β1 with β5 is D62 and D63, two shift nodes whose pairs interleave. The α-helix is T4[25-34], the β3–β5 pairing is P110, and P74 re-phases once. Ubiquitin's mixed sheet uses all three operators. Figure 1 places each node on its contact map.

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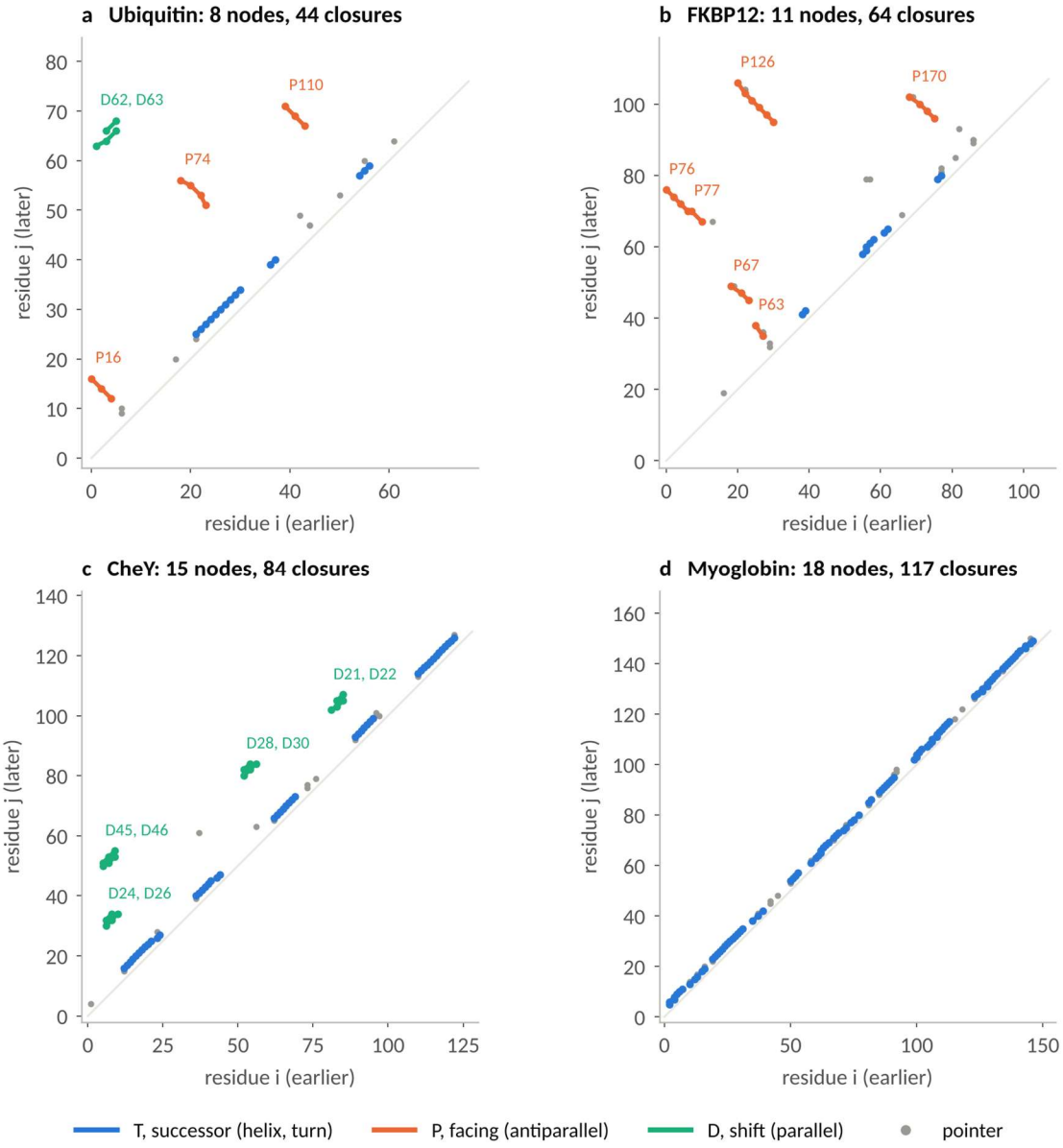


Figure 1. Folds read back to node strings. Contact maps of (a) ubiquitin (1UBQ), (b) FKBP12 (1FKB), (c) CheY (3CHY) and (d) myoglobin (1MBN). Each closure is a point at (earlier residue, later residue). Closures inside a successor node are blue, inside a facing node orange and inside a shift node green; pointers are gray. Facing and shift nodes are labelled by S or d.

Fold classes are node alphabets. Myoglobin, all helix, reads as eighteen successor nodes and no other operator. CheY, a doubly wound α/β fold around a five-stranded parallel sheet, reads in shift and successor nodes with no facing node:

```
D45[5-9] D46[5-9] D24[6-10] D26[6-8] T4[16-25] T3[26-27] T4[40-45] T3[46-47]
D28[52-56] D30[52-54] T4[66-73] D21[81-85] D22[83-85] T4[93-99] T4[114-126]
```

The SH3 domain, all β, reads in facing nodes and one turn:

```
P28[1-5]~-1 P57[2-6] P65[26-28] P83[34-38] T3[51-52]
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FKBP12, a five-stranded antiparallel sheet wrapped around a helix, reads in facing and successor nodes, and its helix reads as T3[58-59] T4[60-62] T3[64-65], the successor operator changing its k inside one helix:

```
P76[0-6] P77[7-10] P67[18-23]~+1 P126[20-30]~-1 P63[25-27]~-1 T3[41-42] T3[58-59]
T4[60-62] T3[64-65] P170[68-75]~+1 T3[79-80]
```

DSSP reads the same closures as n-turns, bridges and ladders and assigns states from their patterns (Kabsch & Sander 1983). The node string reads them as operators whose partners are computed. The pattern is the operator's trace, and the operator is what writes the pattern: the same path read at two locations.

5. The pair word

5.1 Facing and direction

A closure is two operands, i and j. With m bits per residue index, the closure is one word of R_{2m}, the second operand written reversed:

$$W(i, j) = b(i) \parallel J(b(j)).$$

The native facing law acts on this word exactly. Reversed, W(i, j) reads $J(J(b(j))) \parallel J(b(i)) = b(j) \parallel J(b(i)) = W(j, i)$, and W(i, j) is a palindrome only when i = j, so for every closure

$$P(W(i, j)) = W(j, i): \text{ facing swaps the operands.}$$

The ring builder places every word's partner at the facing position, so the two directions of one closure sit at facing addresses: the address of W(j, i) is N − 1 minus the address of W(i, j), with $N = 2^{2m}$. On the diagonal the word is a palindrome and P is complement, so W(i, i) faces W(M − 1 − i, M − 1 − i), with $M = 2^m$: a residue paired with itself faces its complement residue. For every i whose word b(i) is not a palindrome, $W(i, i) = b(i) \parallel J(b(i)) = E(b(i))$. The diagonal of the pair ring is the promotion lane of the residue ring.

The ring's two halves carry the closure's direction. The builder lists the smaller word of each partner pair in rising order in the first half, the listed half, and hangs the partners after them in the second, the created half. W(i, j) with i < j sits in the listed half and W(j, i) in the created half on every ring from R4 to R16, with one exception per ring: the crease swap carries $W(2^m - 2, 2^m - 1)$ and its reverse across the crease. Those two operands are chain neighbours, and chain neighbours never close, since a closure needs $|i - j| > 2$. For every closure a fold can make, the half of the ring is the closure's direction: listed when the earlier residue comes first, created when the later one does. Figure 3a draws W(2, 14), a pair of ubiquitin's hairpin node P16, and its facing word W(14, 2). **RUNS** on every word of R2 through R16.

5.2 The count and the parity sector

In W(i, j) the lowest bit of i and the lowest bit of j sit side by side at the centre of the word, and each operand counts outward from there. A count on i carries leftward from the centre and a count on j carries rightward: the pair word is a two-sided counter with its crease in the middle (Figure 3a).

Each node is a count on this counter. Read step by step through the *E. coli* nodes (Figure 3b), successor nodes step (+1, +1) in 3,775 of 4,109 steps, and the rest step (+2, +2) over a skipped donor. Facing nodes step (+2, −2) in 732 of 947 steps, and shift nodes step (+2, +2) in 353 of 413. A helix counts both operands up by one.

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An antiparallel ladder counts one operand up by two and the other down by two, which is the pleat: every second residue closes across. A parallel ladder counts both up by two. **RUNS.**

The two centre bits carry the register. Their XOR is the parity of i + j, which is also the parity of j − i: the closure's parity sector. A facing node holds S = i + j, a shift node holds d = j − i, and a successor node holds k. A step that changes the held number by an even amount never flips the sector bit, and a step that changes it by an odd amount always does. In the *E. coli* nodes the sector bit flips on none of 5,230 even changes and on all 239 odd changes. The odd changes are the register slips, among them (+2, −3) 93 times and (+3, −2) 89 times in the facing nodes: the β-bulge (Richardson, Getzoff & Richardson 1978) at its operand level. A slip is one bit at the centre of the pair word. Every later pair of the node carries the flipped bit, which is the re-registration measured downstream of each slip in the reading-direction note, where 0.73 of later closures carry the new S and 0.05 return to the old one (Kulik 2026b). **RUNS**: the identity holds on every word, and the counts run on the chains.

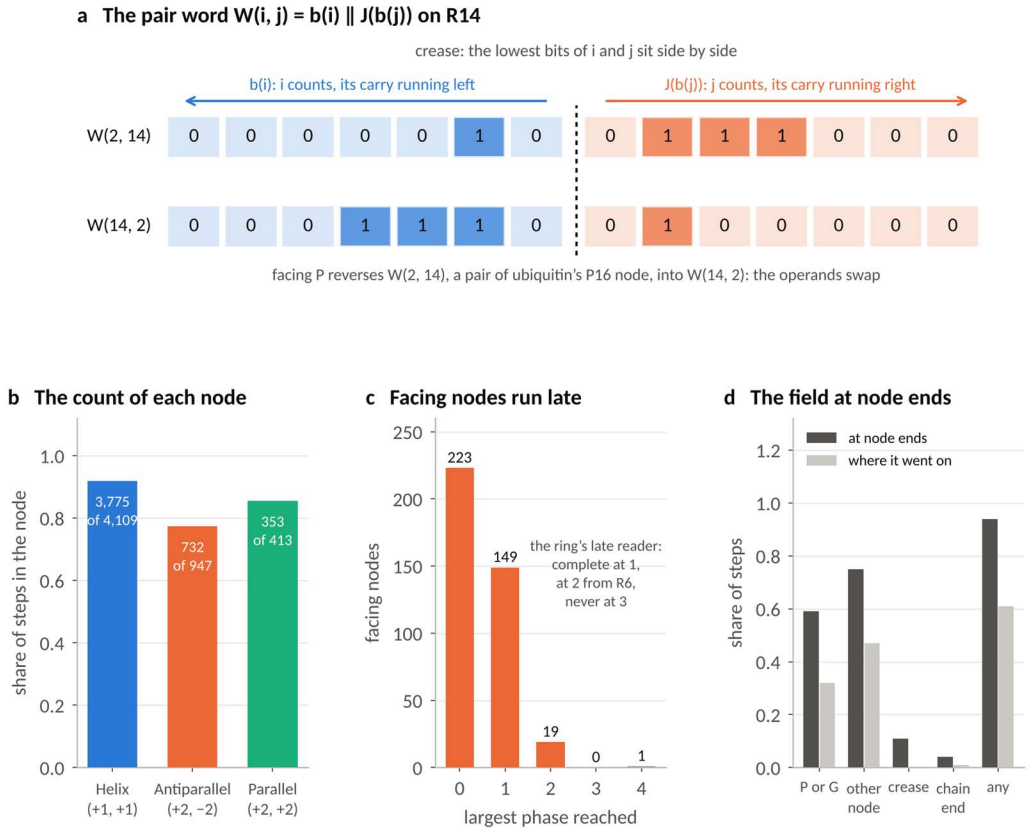


Figure 3. The pair word and the node. (a) W(2, 14) and its facing word W(14, 2) on R14 (m = 7): a pair of ubiquitin's hairpin node P16 and its reverse. The lowest bits of i and j sit at the centre, and each operand's carry runs outward. (b) Share of the steps inside each node type that take its count: (+1, +1) in successor nodes, (+2, −2) in facing nodes, (+2, +2) in shift nodes. (c) Facing nodes by the largest phase they reach. (d) Field criteria at the step past a node's end and at the steps where a node went on: a proline or glycine at the last pair or the next step, another node's boundary within one residue, the hairpin's own crease (facing nodes), the chain end, and any of these.

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5.3 Facing nodes run late

A facing node starts on its crease sum and can re-phase inside the node. By the largest phase each reaches (Figure 3c), 223 of the 392 facing nodes hold S throughout, 149 reach S ± 1 and 19 reach S ± 2; none reaches 3, and one reaches 4 by two steps of two. The compiler paper ran the same depth on the rings: a reader that starts one bit late reads a complete word set on every native ring from R2 to R16, a reader two bits late does so from R6 on, and a reader three or more bits late never does (Kulik 2026a). The facing node meets its partner on time, one late or two late, and does not stand three late. The reader does not carry the path; it lands on the path sooner or later because the path is local on the ring. Lateness stops at the same depth on the ring and on the sheet, two, and the single node at four is the exception the route has to account for. **RUNS** for the phase counts. The route that writes the sheet's phase limit from the ring's late reader is **OPEN**.

6. How a node ends

A node computes its partner and counts its steps. It does not compute where it stops. Three readings of the step past each node's last closure were run, each set against the steps inside nodes where the node went on.

The carry. In residue coordinates, the step past an end crosses a multiple of 2 at 0.83 of ends against 0.70 of inside steps, a multiple of 4 at 0.53 against 0.42, of 8 at 0.34 against 0.31, of 16 at 0.16 against 0.15, and of 32 at 0.07 against 0.07. Split by operator, the end and the inside cross the 2-boundary alike: every step of the T3, facing and shift nodes crosses it, inside and at the end, and T4 nodes cross it at 0.53 inside and 0.50 at the end. The overall excess is the mix of operators, because every node has exactly one end and short nodes weigh more among ends than among inside steps. A carry out of the residue counter is fixed by the operator's step and does not mark the end. **DEAD END** for the end, routed back to its owner, the count.

The field. At the step past the end the field's criteria gather (Figure 3d). A proline or glycine sits at the last pair or the next step, on either strand, at 0.59 of ends against 0.32 inside. Another node's boundary sits within one residue of the next step, the hand-off, at 0.75 against 0.47. A facing node's own hairpin crease, its two strands within nine residues, sits at 0.11 of ends and never inside. The chain end sits at 0.04 against 0.01. At least one criterion sits at 0.94 of ends and at 0.61 of the steps where the node went on. Two providers bind toward the end, the glycine–proline pair and the next node's start, and neither binds it alone. In the terms of a dependency-injection container the end is ambiguous, and the qualifier that separates the 0.94 from the 0.61 is the next readout. **OPEN.**

The pleat. The unbroken stretches of the facing nodes, the runs that hold one S, are read for their span on the first strand. There are 590: 186 single pairs, 192 of span 2, 115 of 4, 56 of 6, 19 of 8, 5 of 10 and 2 of 12, with 15 of odd span; 33 pass span 6, and the longest spans 13. A stretch that has reached span 2 goes on to 4 at 0.51, from 4 to 6 at 0.43, from 6 to 8 at 0.35 and from 8 to 10 at 0.35. Survival falls over the first steps and then holds. There is no wall, no span at which stretches stop, and a counter of fixed width writes a wall at its width. The length of a stretch is not a count. **RUNS.**

The gap. Subtracting what the node owns leaves the end. The partner has its way home: P∘P is the identity, and W(j, i) faces W(i, j). The count has its way home: the step inverts. The end has none in the node's own operands. It is the homeless piece, and the homeless piece is the one that moves: three quarters of ends sit within one residue of another node's boundary. The end is where the next node's start is read. It is the

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second operand of the compiler's 2 + =, arriving from outside the node, an injected dependency. Count, carry and termination are three things with three owners: the count belongs to the node, the carry to the count, and the termination to the field.

7. The gene at the same location

The compiler paper writes each base as a two-bit word, the high bit the ring count (size: purine 0, pyrimidine 1), the low bit the bonding edge, and their XOR the hydrogen-bond grip: A 00, G 01, C 10, T 11 (Kulik 2026a). On one base, P_2 is Watson–Crick pairing (Watson & Crick 1953): A and T face by complement, G and C by reversal. The grip bit is exactly the bit P_2 answers with reversal. The strong pair G–C is the reversal pair, the opposite-spin pair of the occupancy reading, and the weak pair A–T is the complement pair, full facing empty. The base pair and the bond sit on the same two partner pairs of R2. **STANDS**: the double helix supplies the pairs, and the identity is checked on the four bases.

The sixteen two-base boxes of the code are the sixteen words of R4. Their eight P = C words, the class water's slots fall in, are exactly the eight boxes whose two bases share a strength: GG, GC, CG and CC, and AA, AT, TA and TT. The compiler paper's budget rule, which classifies all sixteen boxes, sends the four strong ones to four-codon squares and the four weak ones to split boxes, so inside the P = C class strength alone decides the box. GC (alanine) and CG (arginine) are the palindromes of weight two.

The four anti-palindromes are the words every operator of the ring pairs alike: J, C and P_4 send each one to the same partner. They are GG, glycine; CC, proline; AT, isoleucine with methionine and the start codon ATG; and TA, tyrosine with the stops TAA and TAG. Glycine and proline are the compiler paper's lookahead pair: the backbone twist into a glycine opens to a spread of 0.93 and into a proline clamps to 0.39, against 0.80 overall, one step before the residue itself (Kulik 2026a; Ho & Brasseur 2005). They are also the field criterion that gathers at node ends, 0.59 against 0.32. The start and the stops write the chain's own ends. The four words on which every operator agrees carry the residues that gather at the fold's node ends and the codons that start and stop the gene: the node end and the gene end sit at the same location on R4. **RUNS** for the identities on the standard code. The route that writes a node end from these four words is **OPEN**, and it is the emitter's to write.

The fork bit of the compiler paper, set for the side chains that branch at their first carbon (isoleucine, valine and threonine), is written in the first two bases: purine first, pyrimidine second, the two not both strong, and not ATG. The rule is exact on all 61 sense codons. No fork residue sits on a pyrimidine first base, and no ring-closing residue (phenylalanine, tyrosine, tryptophan, histidine, proline) sits on a purine first base: the size bit of the first base separates the fork route from the ring route. **RUNS** on the 61 sense codons.

The facing law does not own the box class. P_4 keeps the class, square or split, on 12 of the 16 boxes, which a random pairing of the boxes does 18 percent of the time, and complement keeps it on 8. The class belongs to the budget rule and is exchanged on all sixteen boxes by Rumer's flip of bits 1 and 3 (Rumer 1966; Kulik 2026a). **DEAD END** for P_4 as the degeneracy operator, routed to its owner, the budget rule.

Hydropathy takes one step on an open link of the compiler paper. Counting each side chain's water-facing slots, its lone pairs and polar hydrogens, tracks the Kyte–Doolittle scale at r = −0.81 with the slots of nitrogen,

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oxygen and sulfur (Kyte & Doolittle 1982; Kulik 2026a). The compiler paper named sulfur's weak lone pairs as a lane to separate from oxygen's. Counting the slots of nitrogen and oxygen alone gives r = −0.85. **RUNS.**

8. The ledger

Table 2. The read-back's ledger. Every RUNS, STANDS and DEAD END row rests on checks in section 14 of the reproduction notebook.

Result	Status	Section
Water's open slots face their complement in all six orderings; no atom byte does; promotion keeps the class on every lane	RUNS	2.1
R2 as orbital occupancy: P_2 pairs opposite spins (covalent bond, J) and full with empty (hydrogen bond, C)	STANDS	2.2
N–H census: chain 0.77, water 0.10, unlinked 0.13; helices close their own slots, strands hand slots to the field	RUNS	3
Helix start: side chains close the first four N–H at 0.21 against 0.12 in the helix	RUNS	3
Slip residues are closed by the chain at 0.84, as in the kept register	RUNS	3
Water hold: B from 0.99 to 0.74 and water edges from 1.19 to 0.63, one chain edge to four or more	RUNS	3
Node strings: 9.2 nodes per 100 residues, 5.0 closures per node, 0.78 of closures in nodes; fold classes as node alphabets	RUNS	4
Pair word: facing swaps the operands; reverse address N − 1 − a; the diagonal faces its complement and is the promotion lane	RUNS (R2–R16)	5.1
The ring's halves carry the closure's direction; the one exception is two chain neighbours	RUNS (R4–R16)	5.1
Counts inside nodes: (+1, +1) in 3,775 of 4,109 steps, (+2, −2) in 732 of 947, (+2, +2) in 353 of 413	RUNS	5.2
The sector bit is the slip: 0 of 5,230 even changes flip it, 239 of 239 odd changes do	RUNS	5.2
Facing nodes re-phase by 0, 1 or 2, never 3	RUNS	5.3
Route from the ring's late reader to the sheet's phase limit	OPEN	5.3
Carry as the node end	DEAD END (owner: the count)	6
Field criteria at node ends: 0.94 against 0.61 inside; glycine–proline and hand-off both bind toward the end	OPEN	6
Facing stretches survive each pleat step at a steady rate; no wall	RUNS	6

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Result	Status	Section
P_2 on one base is Watson–Crick pairing; grip is the reversal bit	STANDS	7
The R4 anti-palindromes are the glycine, proline, start and stop boxes	RUNS; route to the node end OPEN	7
Fork law on the 61 sense codons	RUNS	7
P_4 as the degeneracy operator	DEAD END (owner: the budget rule)	7
Hydropathy from nitrogen and oxygen slots, r = −0.85	RUNS	7

9. What the emitter must write

Read back to operands, a fold is a node string: about nine operators per hundred residues, each with a type, a parameter and a start, and about a fifth of its closures as pointers. The read-back hands the emitter its specification, and every line of it comes from the sections above.

A node is written by its type, its parameter and its start. The node computes its own partners and counts its own steps, so the emitter does not place closures one by one. Its alphabet is three operators: successor for helices and turns, facing for antiparallel ladders, shift for parallel ones.

A slip is one bit. The emitter writes a re-registration by flipping the sector bit at the centre of the pair word, and every later pair of the node inherits it.

The end is injected. It is not in the node, and the emitter must be handed it by the field. The field's criteria gather at 0.94 of ends, the glycine–proline pair and the next node's start chief among them; the qualifier that separates those ends from the 0.61 of steps where a node goes on is the first readout the emitter needs. On the gene side the glycine and proline boxes are two of the four words every operator pairs alike, and the other two hold the start and two of the stops.

The field's slots stay open for the field. The first four N–H of each helix go to side chains and water, edge strands give their outward N–H to water, and an ordered water's hold grows one step per chain edge. The emitter leaves these slots for the field instead of closing them on the chain.

Pointers are written by address. About a fifth of the closures, 0.23 in the training structures and 0.22 in the *E. coli* chains, carry partners no node computes.

The emitter writes node strings forward from sequence, seals them before any structure is linked, and lands them on chains the program has not read. That run is the landing this read-back does not give, and it is the next paper.

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Data and code

Every number in this paper is printed by the reproduction notebook, NEXUS_Fold_Compiler.ipynb, beside the value printed here. The new results are checks in section 14 under the source label Node Strings; the values carried from the compiler paper and the reading-direction note (the late reader, the lookahead spreads, the slip propagation, the budget rule) are checks in the sections that reproduce those documents, and section 15 collects every check into one ledger. Section 14 reads only the shipped data: data/training_pdb, the 30 training structures; data/ecoli_chains.json, the 155 *E. coli* chains with backbone atoms placed on exact genes; and data/ecoli_water.json, new with this paper. For each chain the water file holds the side-chain nitrogen and oxygen atoms with their gene positions, the crystal waters within 7 Å of them with their B-factors, and the file's median water B: 20,015 waters and 15,841 side-chain atoms in all. provenance/export_water.py rebuilds it from the PDB entries. Figures 1 to 3 plot the values and node strings computed in section 14.

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Technical note on specific runs

Claude (Anthropic), October 2026. These points are kept apart from the author's text. Each names the run or inference it concerns.

1. **The closure rule (sections 3 to 6).** A closure is the nearest backbone O within 3.5 Å of a backbone N, with no hydrogen placed and no angle or energy term. DSSP places hydrogens and scores an electrostatic energy (Kabsch & Sander 1983). The node counts, step counts and census shares are counts under the distance rule; the rates under a DSSP-style rule have not been run.
2. **The node builder (section 4).** Successor nodes allow donor gaps of two; facing and shift nodes allow steps of one to three residues and take pairs with $|i - j| > 5$; every node needs two pairs. The 243 and 1,548 nodes and the 0.77 and 0.78 shares are counts under these settings.
3. **The water census (section 3).** Ordered water depends on resolution and refinement (Carugo & Bordo 1999), and the census does not control for resolution. The B ratio removes each file's B scale through its median water B, not resolution or occupancy. An edge is a contact within 3.5 Å with no angle term, not a verified hydrogen bond. The no-partner class mixes unmodelled water, disordered side chains and N–H with no partner at all.
4. **Identities counted on data (sections 5.1 and 5.2).** The pair-word results are algebra on every word. The sector-bit count, 0 of 5,230 and 239 of 239, follows from the XOR of the two centre bits being the parity of i + j; on the chains it checks the node builder's bookkeeping and counts the slips.
5. **The late-reader link (section 5.3).** The match is between two small counts: the ring's reader stops at two bits late, and the sheet's phases stop at two with one node at four. The paper marks the route OPEN; the match alone does not tie the two.
6. **The base encoding (section 7).** The anti-palindromes of R4 are GG, CC, AT and TA under the compiler paper's encoding and under every encoding in which P_2 is base pairing with A and T on the palindromes 00 and 11. With G and C on the palindromes they would be GC, CG, AA and TT. The encoding was fixed in the compiler paper by the size and edge bits, before these runs.
7. **Stops and variant codes (section 7).** Two of the three stops sit in TA; the third, TGA, sits in TG, which is not a P = C word. The fork rule is exact on the standard code; in the vertebrate mitochondrial code ATA reads methionine, and the rule as written classes it as a fork.
8. **The P = C boxes (section 7).** The square and split classes inside the P = C words follow from the compiler paper's budget rule; they are not an independent result.
9. **Data already read (all sections).** The training structures and the *E. coli* chains were read in earlier rounds of this program. No result here is a sealed landing; the sealed test of the node string is the emitter's.
10. **A correction carried from the running audit.** Entry 3 of the running audit reported a knee in pleat survival (0.55, 0.44, 0.26) from pairs collected in windows. With the node runs used here, survival is 0.51, 0.43, 0.35 and 0.35, and there is no knee; entry 4 of the audit records the correction. The census in section 3 also replaces the audit's earlier census from the raw downloads (19,202 N–H) with the census of the shipped water file.
11. **Residue numbering.** Residue numbers in node strings count from 0 along the model chain and are not PDB residue numbers.

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